LogicPath

Product Requirements Specifications

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1 Introduction

Sound reasoning and critical thinking are foundational to academic achievement, civic participation, and professional decision making. The ability to analyze claims, detect flawed arguments, and compare competing sources of information directly influences how individuals vote, make financial decisions, pursue education, and engage in public discourse. In high information and digitally embedded societies, reasoning is not a specialized academic skill but a core ability that supports informed decision making and responsible autonomy. When reasoning skills decline, the consequences extend beyond individual performance to broader social trust, democratic processes, and institutional credibility (Butler 2024).

In our modern society the ability to think critically, judge evidence, and logically reason, has been an increasingly difficult challenge. The rapid growth of digital media, algorithm driven content, and the increasing normalization of misinformation have created an environment in which people are frequently required to assess claims, sources, and arguments without the analytical tools required to do so. Educational researchers have identified this gap as a pressing societal issue, especially since traditional educational models struggle to adapt to the pace and format of modern information consumption. A 2019 study found that 96 percent of students were unable to reliably evaluate the credibility of online sources, which highlights a significant systemic weakness in digital and informational literacy (Breakstone et al. 2019) More recently, the Organization for Economic Cooperation and Development (OECD) reported a decade long stagnation in adult problem-solving and reasoning skills, despite increasing educational attainment and access to information (OECD, 2024). Together, these findings suggest that exposure to information alone is insufficient. And that structured reasoning skills must be deliberately cultivated.

The core problem is not a lack of intelligence or motivation, by rather a lack of accessible and engaging systems that teach logical reasoning as a transferable and practiced skill rather than a purely academic abstraction. Logic instruction is often isolated within philosophy or mathematics courses, framed in highly formal language, and disconnected from everyday decision making. As a result, many learners fail to recognize its relevance, leading to disengagement and shallow retention. An effective solution must satisfy a few key characteristics. It must lower barriers to entry, connect formal reasoning to practical real-world context, adapt to individual learner ability, and sustain engagement over time.

LogicPath is designed to meet these needs by providing a modern, interactive, and accessible platform for developing strong reasoning skills. Rather than treating logic as an abstract academic subject, LogicPath reframes reasoning as a practical tool for everyday use that can be strengthened through guided practice. The platform integrates informal and formal logic instruction through interactive modules, adaptive learning algorithms, and gamified progression systems that encourage consistent engagement. Users receive immediate feedback on their reasoning, allowing them to identify common fallacies, evaluate evidence, and build confidence in constructing sound arguments.

By incorporating real-world scenarios drawn from news, debates, and everyday decision-making, LogicPath bridges the gap between theoretical concepts and practical application. Adaptive difficulty ensures that learners are continually challenged at an appropriate level, while gamified elements such as streaks, achievements, and progression paths help sustain long-term motivation. Through this approach, LogicPath aims to cultivate durable reasoning habits that extend beyond the platform and into academic, professional, and personal environments.

The scope of LogicPath is defined around high school juniors and seniors preparing for standardized tests such as the SAT and ACT, college students in philosophy, law, or STEM programs with formal logic requirements, and anyone seeking to strengthen critical thinking skills. This focus reflects the alignment with the platform's gamified design. Gaming audiences tend to skew younger, LogicPath leverages structured progression systems, quest-based learning, and adaptive difficulty engaging learners during formative academic years while still remaining relevant to adult professionals. The platform is therefore positioned not as a universal education replacement, but as a supplemental reasoning tool tailored to learners who benefit from structured logic development tied to measurable academic and professional goals.

Ultimately, LogicPath addresses the shortcomings of traditional logic education by offering a scalable, technology-driven solution that is both relevant and engaging. In an era defined by information overload and widespread misinformation, LogicPath provides learners with the tools they need to navigate complex information environments with clarity, confidence, and intellectual rigor.

2 Product Description

LogicPath is a web-based educational system composed of two primary components: the learner-facing application and the backend server infrastructure. The learner-facing application delivers interactive logic lessons, gamified challenges, and real-world reasoning exercises through a modern, browser-based interface. The backend infrastructure manages user authentication, adaptive learning algorithms, performance analytics, and long-term data storage. Together, these components form an integrated platform designed to teach logic as a practical, transferable skill rather than an abstract academic subject.

When a learner accesses LogicPath through a web browser, the frontend application retrieves personalized learning data from the server and presents the user with modules tailored to their current skill level. As the learner progresses through lessons, completes challenges, or interacts with real-world reasoning tasks, the application sends performance data to the server. The backend processes this data through adaptive learning engines that adjust difficulty, recommend targeted practice, and update the learner's progression path. Updated feedback and learning recommendations are then returned to the frontend, ensuring that each user receives a dynamic and individualized learning experience.

LogicPath's design emphasizes accessibility and engagement. The system incorporates visual aids, interactive diagrams, and simulations to simplify abstract concepts such as argument structure, logical operators, and fallacy identification. Gamification elements, including streaks, quests, and achievements, encourage consistent participation and help learners build long-term reasoning habits. Real-world application exercises connect

logic to news articles, debates, and everyday decision-making scenarios, reinforcing the relevance of critical thinking skills beyond the platform.

2.1 Key Product Features and Capabilities

2.2 Interactive Module

Lessons progress from informal reasoning to formal logic, allowing learners to build foundational skills before advancing to symbolic logic, argument mapping, and structured problem solving.

2.3 Gamified Learning

Quests, streaks, badges, and achievements motivate sustained engagement and transform logic practice into a rewarding, game-like experience.

2.4 Real-World Application

Real-World Application are exercises that appear weekly with contemporary examples drawn from news media, public discourse, and practical decision-making scenarios. This bridges the gap between theoretical logic and applied reasoning using contextual scenarios.

2.5 Adaptive Learning

An adaptive logic engine which dynamically adjusts content difficulty based on the user performance metrics such as accuracy rate. This prevents cognitive overload or stagnation while providing an individualized progression.

2.6 Visual Aids

The platform will integrate diagrams, logical flowcharts, argument maps, and interactive simulations to reduce abstraction when learning formal logic.

2.7 Major Components (Hardware/Software)

2.7.1 Hardware: LogicPath requires minimal hardware from the end user. Any modern device capable of running a modern web browser (laptop, desktop, tablet, or smartphone) can access the platform. On the back end, cloud-hosted servers provide computational resources for adaptive learning algorithms, user authentication, and data processing.

Operating System: Windows, macOS, Linux (Any Modern OS)
Web browser: Chrome, Firefox, Edge, Safari (Any Modern Web Browser)

2.7.2 Software Architecture:

2.7.2.1 Presentation Layer: Built using React, HTML, and CSS.
Provides the user interface for lessons, challenges, visualizations, and feedback.
Handles navigation, module interaction, and real-time update

2.7.2.2 Application Layer: The Application Layer is implemented using Spring Boot, a Java-based framework designed for building scalable, modular, and production-ready backend services. This layer contains the core logic that powers the LogicPath platform, including:

- **Learning Module Engine**
  Delivers structured lessons, evaluates responses, and manages module progression.

- **Gamification Engine**
  Handles streaks, achievements, quests, and reward logic.

- **Adaptive Learning Engine**
  Analyzes user performance and adjusts difficulty levels dynamically.

- **Feedback and Analytics Engine**
  Processes user data to generate personalized recommendations and insights.

  Spring Boot enables LogicPath to maintain a clean microservice-ready architecture, support RESTful API communication with the frontend, and integrate seamlessly with MySQL. Its built-in dependency management, security modules, and scalability make it well-suited for an educational platform expected to grow over time.

2.7.2.3 Data Layer: Powered by MySQL Stores:

- User profiles

- Learning history

- Performance metrics

- Progression data

- Provides strong relational integrity, fast query performance, and broad compatibility with Spring Boot supports future scaling as the userbase grows

2.7.2.4 Tech Stack

- **Frontend:** React, HTML, CSS
- **Backend:** Spring Boot (Java)
- **Database:** MySQL
- **Hosting:** Cloud-based web server and database services

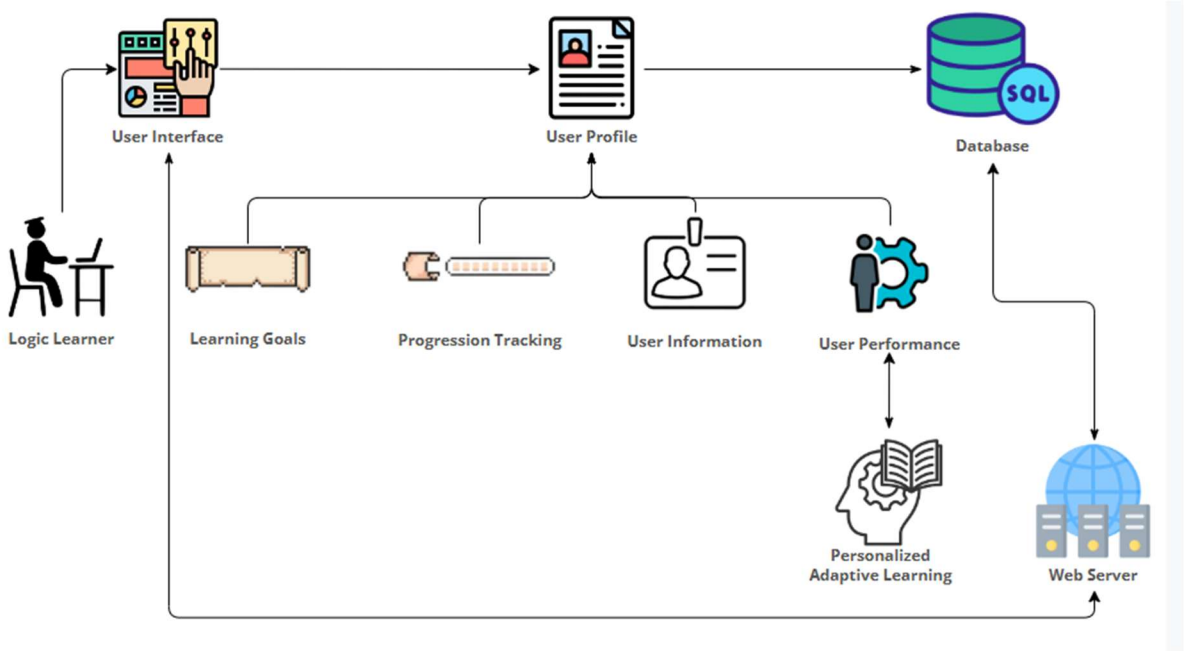

Figure 1

3 Product Requirements

3.1.1 Functional Requirements

3.1.1.1 Progress Tracking (O: Paul)

Algorithm Requirements:

- FR-1.1: The system shall increase the learner's XP by a fixed amount defined per exercise type upon successful completion of each exercise.

- FR-1.2: The system shall increase the learner's level by exactly 1 when the learner's accumulated XP meets or exceeds the XP threshold defined for that level.
- FR-1.3: The system shall unlock new quests or modules when the learner's level meets or exceeds the minimum level threshold associated with that content.

Processing Requirement:

- FR-1.4: The system shall award special achievements when the learner reaches a predefined level milestone (e.g., levels 5, 10, 25).

(**M: Trent 4/25/2026):** "sufficient amount" replaced with specific threshold language, IDs added, format fixed.

3.1.1.2 Adaptive Difficulty Adjustment (O: Trent)

Algorithm Requirements:

- FR-2.1: The system shall automatically adjust the difficulty level of the next assigned question based on the user's most recent attempt result.
- FR-2.2: The system shall not adjust difficulty by more than one level above or below the user's current difficulty tier per adjustment cycle.
- FR-2.3: The system shall evaluate the adjusted difficulty against defined threshold criteria before selecting and assigning the next question.
- FR-2.4: The system shall assign a question at the default "Beginner" difficulty tier when the user has no prior attempt history.

Processing Requirements:

- FR-2.5: The system shall assign a new question to the user's active session within 2 seconds of submission.
- FR-2.6: The system shall notify the user and assign a question from the nearest available difficulty tier when no questions exist at the determined tier.

(**M: Trent 4/25/2026):** IDs added, Algorithm/Processing labels added, FR-2.5 moved to Processing.

3.1.1.3 Instructor Module/Question Assignment (O: Caleb)

UI Requirements:

- FR-3.1: The system shall display a list of available learning modules to authenticated instructor users.
- FR-3.2: The system shall allow instructors to search and filter available modules and questions by criteria including topic and difficulty level.
- FR-3.3: The system shall display the questions contained within a selected module when an instructor selects that module.
- FR-3.4: The system shall display a confirmation summary of selected modules or questions and all assignment settings before the assignment is finalized.

Processing Requirements:

- FR-3.5: The system shall allow instructors to select one or more modules for assignment.
- FR-3.6: The system shall allow instructors to select and assign individual questions within a module.
- FR-3.7: The system shall allow instructors to assign selected modules or questions to a specific class or group of students.
- FR-3.8: The system shall allow instructors to specify a start date for when an assignment becomes available to students, or to make the assignment available immediately.
- FR-3.9: The system shall allow instructors to specify an end date after which students may no longer submit responses to an assignment.
- FR-3.10: The system shall allow students to access modules and questions that have been assigned to them by their instructor.

(**M: Trent 4/25/2026):** "fuestions" typo fixed, IDs added, split into UI/Processing categories.

3.1.1.4 Immediate Feedback (O: Krishna)

Algorithm Requirement:
- FR-4.1: The system shall evaluate the learner's submitted answer for each logic challenge using predefined solution rules.

UI Requirements:
- FR-4.2: The system shall display one of three feedback states — Correct, Incorrect, or Partially Correct — upon submission of each answer.
- FR-4.3: The system shall provide a brief explanation of 1–3 sentences describing the correct reasoning steps for the challenge.
- FR-4.4: The system shall allow the learner to retry the challenge after receiving feedback.

Data Requirement:
- FR-4.5: The system shall record the learner's performance result (Correct, Incorrect, or Partial) for progress tracking purposes.

(**M: Trent 4/25/2026):** IDs added, split into UI/Algorithm/Data categories, label corrected from "Data Requirements" to the proper three-way split.

3.1.2 Non-Functional Requirements

3.1.2.1 User Data Encryption (O: Trent)

- NFR-1.1: The system shall encrypt all sensitive user data at rest using AES-256 encryption.
- NFR-1.2: The system shall use TLS 1.2 or higher for all data exchanged between the frontend and backend over the network.

(**M: Trent 4/25/2026):** AES-256 and TLS 1.2 specifics added.

3.1.2.2 Availability (O: Paul)

- NFR-2.1: The system shall maintain a minimum uptime of 99% measured over any rolling 30-day period, excluding scheduled maintenance windows.
- NFR-2.2: The system shall provide access to core learning functionality — including lessons, exercises, and progress tracking — throughout any active authenticated user session.

(**M: Trent 4/25/2026):** 99% uptime over rolling 30-day period added.

3.1.2.3 Performance (O: Krishna)

- NFR-3.1: The system shall load core user interface components within 5 seconds under a stable network connection.

(**M: Trent 4/25/2026):** Wording changed

3.1.2.4 Scalability (O: Caleb)

- NFR-4.1: The system shall support the addition of new learning modules and achievements without requiring changes to the existing system architecture.

(**M: Trent 4/25/2026):** Wording changed

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4 Identification of Case Study

LogicPath is being developed for high school juniors and seniors preparing for standardized tests, college students enrolled in logic-heavy disciplines such as philosophy, law, and computer science, and adult learners seeking to sharpen their critical thinking for career advancement. As discussed, these users currently lack a single resource that is both comprehensive in its coverage of formal and informal logic and engaging enough to sustain long-term motivation. LogicPath addresses this gap by combining structured, progressive learning modules with gamification elements, such as quests, experience points, streaks, and boss battles, to create a dynamic and rewarding educational experience.

The primary users of LogicPath are students who must navigate logic-intensive coursework or standardized testing with little structured guidance and few interactive tools to support them. These learners often turn to scattered YouTube videos, dry textbooks, or disconnected practice problems, with no reliable way to verify their understanding or track their growth over time. LogicPath will give them a clear learning path, immediate feedback on exercises, and a visual record of their progress empowering them to take ownership of their development in a way current resources do not allow.

To test and evaluate the effectiveness of LogicPath, a case study will be conducted using undergraduate students as a beta group. Participants will be recruited from courses with logic or critical thinking components, such as introductory philosophy or discrete mathematics. These students will use LogicPath as a supplemental learning tool over the course of a semester, progressing through modules at their own pace while the platform's adaptive learning engine tracks their performance, accuracy, time spent, and module completion rates. Participants will be given opportunities to submit ongoing feedback through in-app prompts and will complete a structured evaluation at the end of the study period. Analytics collected throughout the case study will be reviewed to assess whether LogicPath meaningfully improves students' engagement with logic material and their self-reported confidence in reasoning skills.

As the platform grows, feedback from users, educators, and institutional administrators will become central to expanding and refining its content. It may be necessary to customize certain modules or difficulty settings to align with specific course requirements at different institutions. Future expansion of the case study could extend to high school classrooms, lifelong learner communities, or professional development programs seeking to improve workforce critical thinking skills.

5 Glossary

- **Logic:** the systematic use of symbolic and mathematical techniques to determine the forms of valid deductive argument.
- **Formal/Informal Logic:** Formal logic is based off deductively valid reasoning. Informal logic is based off natural languages.
- **IDE**: Integrated Development Environment
- **CI**: Continuous Integration
- **CD**: Continuous Deployment
- **AES-256:** Advanced Encryption Standard with a 256-bit key, used for encrypting data at rest.
- **TLS:** Transport Layer Security, a cryptographic protocol used to secure network communications.
- **FR:** Functional Requirement
- **NFR:** Non-Functional Requirement
- **O:** Originator, the team member who authored the requirement.
- **M:** Modifier, the team member who last revised the requirement (to be assigned in future revisions).

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